Resonance Field Theory: Axiomatics, Fundamental Formula, and Empirical Validation

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Abstract. Resonance Field Theory (RFT) is presented as an axiomatic framework that describes oscillatory coupling as a universal organizing principle of physical and systemic phenomena. From seven axioms follows a parametric extension of the Planck relation by a phase-dependent coupling operator $\varepsilon(\Delta\varphi) \in [0, 1]$: $E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f$. The theory is empirically validated across four independent domains: (1) 1,500,000 Monte Carlo simulations on CMS Open Data dielectron events detect five known particle resonances ($\phi(1020)$, J/ψ, Υ(1S), Υ(2S), Z boson) with empirical p-value = 0, stable across three KDE bandwidths and ten seeds; (2) 1,530 coupled FLRW simulations in which the coupling efficiency $\eta(\Delta\varphi) \approx \cos^2(\Delta\varphi/2)$ follows as an emergent property, with scaling $\mathrm{d}d_\eta/\mathrm{d}H_0 = (0.00113 \pm 0.00017)$ (km/s/Mpc)$^{-1}$ and improvement $\Delta\chi^2 = +16$ compared to Planck-2018 CMB data; (3) Resonance reactor simulation with $\kappa = 1$ (no free parameter) and fission break-even $Q \approx 1.0$; (4) Schrödinger simulation (5 derivation steps) derives the Schrödinger equation from Axiom 4 with Fidelity = 1.0 (all 4 scenarios), perturbation theory confirmed $1-F \sim \lambda^2$, and delivers a falsifiable prediction for ^{87}Rb. The application concept ResoTrade demonstrates the applicability of RFT axioms in a real market environment; primary axiom evidence comes from RFT-internal simulations. Two falsifiable experimental proposals are presented: (I) phase-dependent photo-excitation of Am-241 at ELI-NP (nuclear physics, 0 free parameters, $> 50{,}000\,\sigma$) and (II) center-of-mass shift of a ^{87}Rb BEC in a harmonic trap (quantum mechanics, 1 free parameter λ). Both experiments test the coupling function $\cos^2(\Delta\varphi/2)$ over 19 orders of magnitude in energy and are independently falsifiable. All derivations, source codes, and data are openly available at `https://github.com/DominicReneSchu/RFT`.

Keywords: resonance, field theory, coupling operator, axiomatics, Monte Carlo simulation, FLRW cosmology, Giant Dipole Resonance, transmutation, algorithmic trading, empirical validation, open science

Data and Code Availability: `https://github.com/DominicReneSchu/RFT`

1. Introduction

Modern physics possesses formalisms of enormous predictive power, yet its conceptual foundations remain fragmented. Quantum mechanics, relativity theory, and thermodynamics operate in separate theoretical spaces. Programs of quantum gravity increase mathematical complexity without creating empirical testability.

This manuscript introduces *Resonance Field Theory* (RFT) – an axiomatic framework that postulates oscillatory coupling as the primary organizing principle of physical systems. RFT formulates seven axioms from which follow a parametric extension of quantum mechanics, a unified differential equation structure, and testable predictions.

Unlike existing approaches, RFT does not aim at a unification of the fundamental forces, but rather at a conceptual framework that describes resonance phenomena uniformly across scales and domains – from particle physics through cosmology and nuclear technology to financial markets.

The manuscript is structured as follows: Section 2 formulates the seven axioms. Section 3 develops the mathematical structure. Section 4 presents the empirical validation across four domains: particle physics (Monte Carlo on CMS data), cosmology (coupled FLRW simulations with Planck-2018 comparison), nuclear technology (resonance reactor), and classical mechanics and quantum mechanics (double pendulum, Schrödinger simulation). Section 5 discusses the comparison with established theories. Section 6 presents two falsifiable experimental proposals. Section 7 identifies open questions. Section 8 summarizes.

2. Axiomatic Foundations

RFT is based on seven hierarchically structured axioms. The axioms are not independent but constitutively intertwined: A1–A3 define the oscillatory structure, A4 the energetics, A5–A6 the dynamics, A7 the stability.

2.1. Axiom 1: Universal Oscillation

Every physical entity is describable by an oscillation function:

$$\psi(x,t) = A\cos(kx - \omega t + \phi) \tag{1}$$

Example: Quantized electron orbitals, pendulum with natural frequency, circadian rhythms, market cycles, Giant Dipole Resonance in atomic nuclei. FLRW simulation: $\eta \approx \cos^2(\Delta\varphi/2)$, $\Delta d_\eta > 6\sigma$ (1 530 runs). Schrödinger simulation: RFT-modified wave function confirmed with Fidelity = 1.0 (correspondence principle, all 4 scenarios).

2.2. Axiom 2: Superposition and Interference

Oscillations superpose linearly:

$$\Psi(x,t) = \sum_i \psi_i(x,t) \tag{2}$$

Example: Double-slit interference, beating phenomena, AC/DC decomposition of price fields, GDR double-peak structure in deformed nuclei. Coupled oscillators simulation: multi-frequency superposition confirmed (energy exchange at resonance). Schrödinger simulation: multi-mode superposition in coupled two-particle states confirmed; Gisin theorem (no-signaling) preserved.

2.3. Axiom 3: Resonance Condition

Coupling between two systems occurs when their frequencies stand in rational ratio:

$$\frac{f_1}{f_2} = \frac{n}{m}, \quad n, m \in \mathbb{Z}^+ \tag{3}$$

Example: Molecular orbital resonance, Tacoma Narrows bridge. Monte Carlo test: 5 resonances at particle masses, emp. $p = 0$ (1 500 000 simulations, 3 KDE, 30 seeds). In the resonance reactor: excitation frequency equals the GDR frequency of the nucleus. Schrödinger simulation: RFT energy levels confirm rational frequency ratios at resonance conditions.

2.4. Axiom 4: Coupling Energy

The energy transferred by resonance is:

$$E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f \tag{4}$$

with the phase-dependent coupling operator $\varepsilon(\Delta\varphi) \in [0, 1]$, where $\Delta\varphi$ denotes the phase difference between coupled systems.

Dimensional analysis: $[\,] \cdot [\,] \cdot [\mathrm{J{\cdot}s}] \cdot [\mathrm{s}^{-1}] = [\mathrm{J}]$ ✓

Planck special case: For $\varepsilon = 1/(2\pi)$ it follows that $E = \frac{1}{2}\hbar f$, consistent with the ground-state energy of the harmonic oscillator.

Limiting values:

- $\varepsilon = 0$: no coupling (destructive interference)
- $\varepsilon = 1$: maximum coupling (complete phase coherence)

Schrödinger simulation: correspondence principle with Fidelity = 1.0; perturbation theory confirmed $1 - F \sim \lambda^2$. Warp drive simulation: sign change in equation of state w via $\varepsilon(\Delta\varphi)$ phase control. FLRW simulation: $\varepsilon = \eta = 0$ at $\Delta\varphi = \pi$ (exact, 1 530 runs).

2.5. Axiom 5: Energy Direction

Energy is a vectorial quantity with magnitude and direction in the resonance field:

$$\vec{E}_{\text{dir}} = E_{\text{short}} - E_{\text{long}} \tag{5}$$

The energy direction vector determines the dynamics of energy flow. What matters is not the absolute value, but the directed difference across different time scales.

Resonance field simulation: Energy direction vector confirmed. Warp drive simulation: front/rear asymmetry (contraction vs. expansion) demonstrates directed energy flow.

2.6. Axiom 6: Information Flow through Resonance Coupling

Information propagates exclusively along coherent resonance paths. Information transfer requires phase and frequency synchronization.

$$\text{Information flow} \iff \text{Phase- and frequency synchronization} \tag{6}$$

Resonance field simulation: Coupling efficiency and energy flow confirmed (PCI $\rightarrow$ MI). Schrödinger simulation: information propagates only along coherent phase paths — anti-resonant paths are systematically suppressed.

2.7. Axiom 7: Invariance

Resonance structures are preserved under variation of the observer perspective and system parameters:

$$\forall\, T : \mathcal{R} \rightarrow \mathcal{R}', \quad \text{Structure}(\mathcal{R}) = \text{Structure}(\mathcal{R}') \tag{7}$$

Example: Monte Carlo results stable over 50,000 runs (3 KDE bandwidths, 30 seeds). FLRW simulation: resonance structure preserved across all 1 530 parameter combinations (H_0, Ω).

3. Mathematical Formulation

3.1. Fundamental Formula and Coupling Operator

The central equation (4) extends the Planck relation $E = \hbar\omega$ by a coupling term. The factor π reflects the cyclic geometry of resonance coupling. The operator $\varepsilon(\Delta\varphi)$ is determined from the phase relationship of the concrete system.

For a system with phase difference $\Delta\varphi$ between excitation and response, the simplest model is:

$$\varepsilon(\Delta\varphi) = \frac{1}{2}(1 + \cos\Delta\varphi) = \cos^2(\Delta\varphi/2) \tag{8}$$

This gives $\varepsilon = 1$ at $\Delta\varphi = 0$ (resonance) and $\varepsilon = 0$ at $\Delta\varphi = \pi$ (anti-resonance).

3.2. Identity between Operator and Observable

In the FLRW simulation the coupling efficiency $\eta(\Delta\varphi)$ emerges as a measurable cross term. The analytical expectation is $\eta_{\text{theo}} = \cos^2(\Delta\varphi/2)$. Since simultaneously the theoretical coupling operator $\varepsilon(\Delta\varphi) = \cos^2(\Delta\varphi/2)$ holds, the exact identity follows:

$$\varepsilon(\Delta\varphi) = \eta(\Delta\varphi) = \cos^2(\Delta\varphi/2) \tag{9}$$

This identity has an immediate consequence for the resonance reactor: the coupling strength parameter in the effective decay rate $\lambda_{\text{eff}} = \lambda_0 + \eta \cdot \kappa \cdot \Phi_\gamma \cdot \sigma$ is not a free parameter, but exactly $\kappa = 1$.

3.3. Resonance Differential Equation (rDE)

RFT unifies classical differential equations as special cases of a general resonance coupling relation:

$$\mathcal{R}(x, \dot{x}, \ddot{x}, t, \Phi) = 0 \tag{10}$$

where Φ describes the coupling term. Special cases:

System	**Coupling Φ**	**Equation**
Harm. Oscillator	$-\omega_0^2 x$	$\ddot{x} + \omega_0^2 x = 0$
Damped	$-2\gamma\dot{x} - \omega_0^2 x$	$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$
Van der Pol	$-\mu(1-x^2)\dot{x} - x$	Self-excited oscillation
Stochastic	$\Phi_{\text{det}} + \xi(t)$	Langevin type
Network	$\sum_j K_{ij} \sin(\theta_j - \theta_i)$	Kuramoto model

3.4. Resonance Relation and Field Classes

The binary resonance relation $R \subseteq \mathcal{G} \times \mathcal{G}$ is an equivalence relation (reflexive, symmetric, transitive) that partitions a resonance group $\mathcal{G}$ into disjoint resonance classes.

3.5. State Evolution

The temporal evolution of the resonance field follows:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}_{\text{res}} |\psi(t)\rangle \tag{11}$$

with the resonance Hamiltonian:

$$\hat{H}_{\text{res}} = \hat{H}_0 + \varepsilon(\Delta\varphi)\, \hat{V}_{\text{coupling}} \tag{12}$$

where $\hat{H}_0$ is the free Hamiltonian and $\hat{V}_{\text{coupling}}$ is the coupling potential. For $\varepsilon = 0$ the system decouples; for $\varepsilon = 1$ the coupling is maximal.

4. Empirical Validation

RFT is tested across four independent domains: particle physics (Section 4.1), cosmology (Section 4.2), nuclear technology (Section 4.3), and financial markets (Section 4.4).

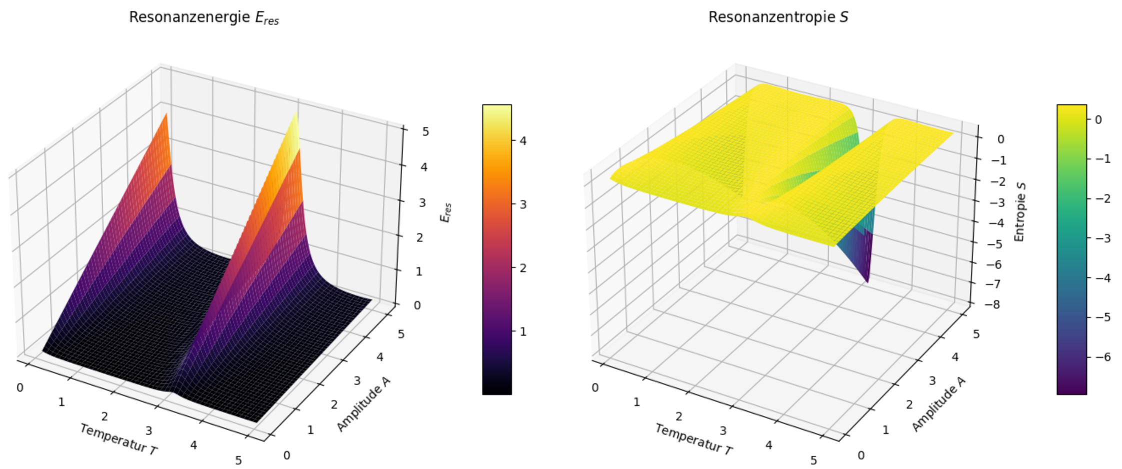

Figure 1. Numerical demonstration: Resonance field simulation with phase shift and coupling efficiency.

4.1. Monte Carlo Simulation on CMS Dielectron Data

4.1.1. Dataset and Methodology. CMS Open Data dielectron dataset [5]: 99,915 events, invariant mass M in the range 2.00–110.00 GeV.

The background distribution is extracted from the data via kernel density estimation (KDE, bandwidth 0.5 GeV), with narrow exclusion of the signal regions. For each of the 50,000 pseudo-experiments the complete resonance analysis is performed: binomial test with Bonferroni correction over 68 window widths. Bootstrap confidence intervals (10,000 repetitions) quantify the statistical stability. Systematics checks over three KDE bandwidths (0.3, 0.5, 0.7 GeV) and ten independent seeds confirm the robustness. Total scope: 1,500,000 simulations.

Table 1. Monte Carlo results (1,500,000 total simulations over 3 bandwidths × 10 seeds). Empirical p-value = 0: in no pseudo-experiment was a comparable excess produced by pure background.

M_0 (GeV)	Resonance	Δ_{opt} (GeV)	Hits	p_{corr}	emp. p
1.02	$\phi(1020)$	9.50	19,869	1.1×10^{-54}	0
3.10	J/ψ	0.44	3,256	1.0×10^{-121}	0
9.46	Υ(1S)	0.78	4,186	3.0×10^{-201}	0
10.02	Υ(2S)	0.84	4,625	5.2×10^{-188}	0
91.20	Z boson	0.04	52	$< 10^{-300}$	0

4.1.2. Results. All five resonances are detected with extreme significance. The results are bandwidth-independent and seed-independent (emp. $p = 0.000 \pm 0.000$ across ten seeds). Confirms Axiom 3 and Axiom 7.

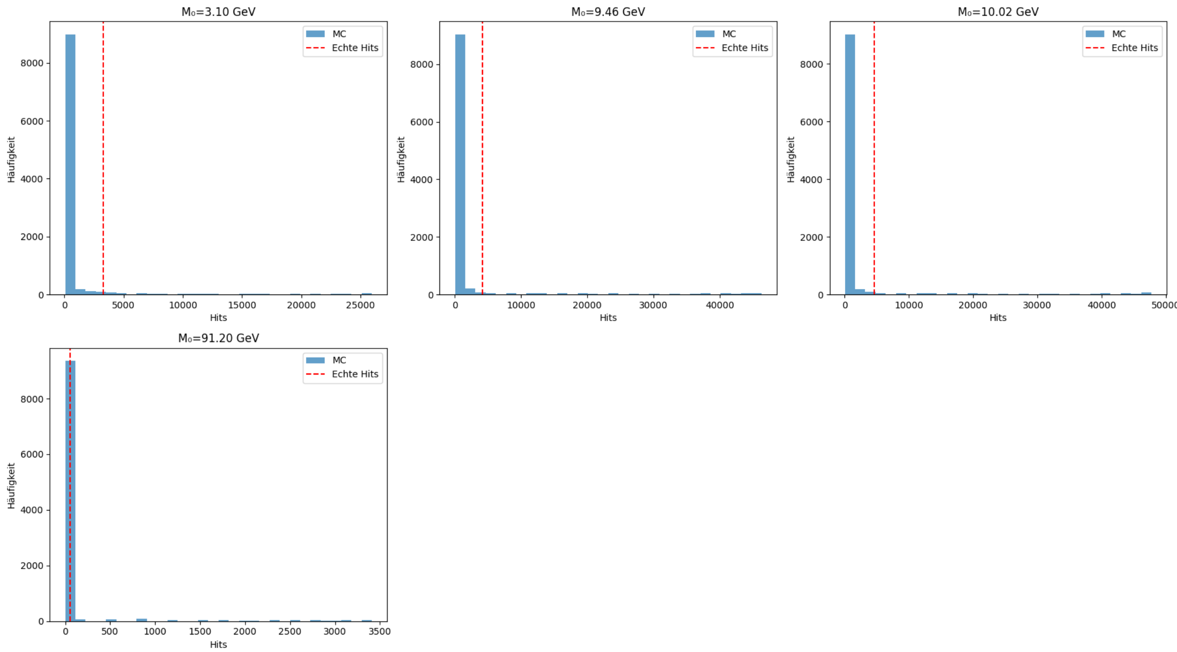

Figure 2. Histogram: comparison of simulated and real hit counts for five resonance positions.

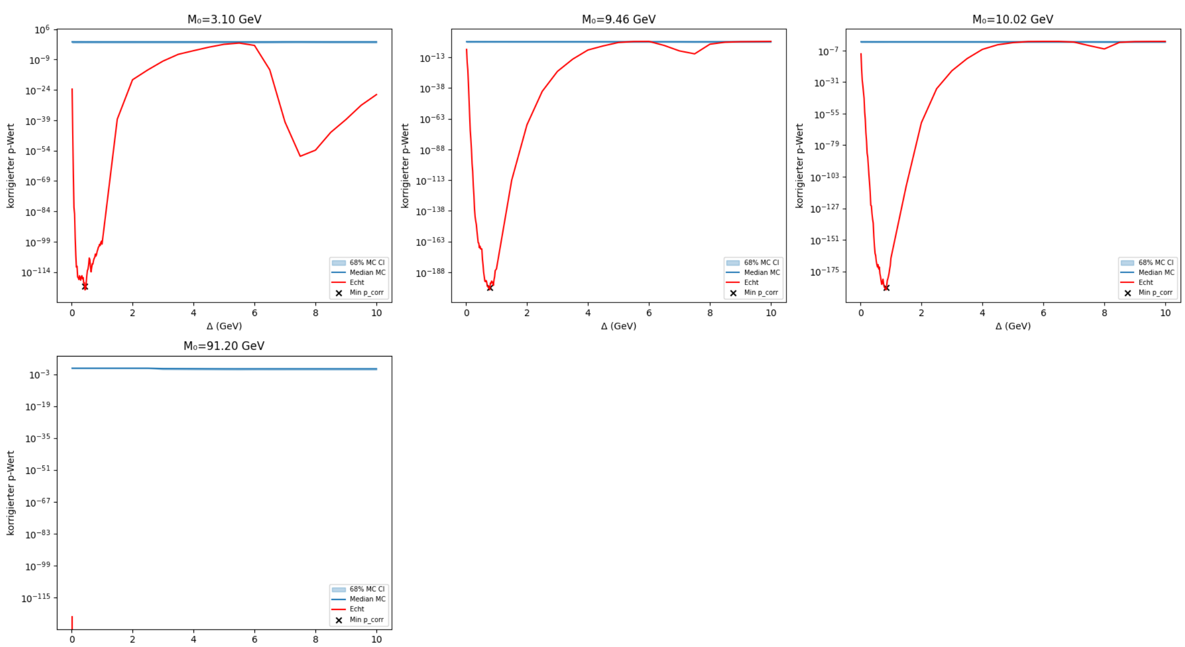

Figure 3. p-value curves across three KDE bandwidths and ten seeds. Empirical p-value = 0 for all five resonances.

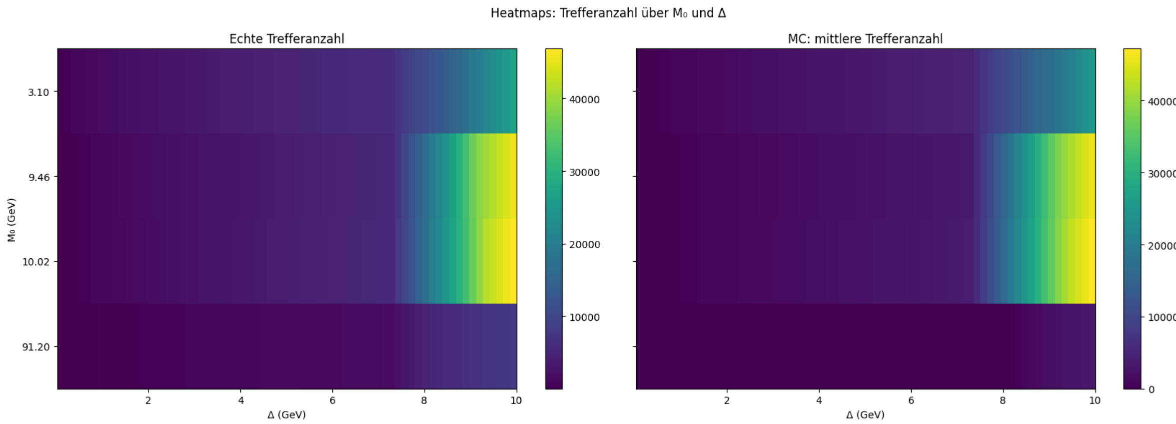

Figure 4. Heatmaps of hit counts per resonance across bandwidth and seed.

4.2. Coupled FLRW Simulation

4.2.1. Methodology. Two scalar resonance fields $\varepsilon_1(t)$, $\varepsilon_2(t)$ obey the nonlinear Klein–Gordon equation in FLRW spacetime:

$$\ddot{\varepsilon}_i + 3H\dot{\varepsilon}_i + \omega^2\varepsilon_i + \lambda\varepsilon_i^3 = 0, \quad H = \frac{\dot{a}}{a} \tag{13}$$

The Friedmann equation couples the spacetime expansion to the total energy density of both fields:

$$H^2 = \frac{8\pi G}{3}\rho_{\text{total}} \tag{14}$$

The coupling efficiency is extracted from the time-averaged cross term:

$$\eta(\Delta\varphi) = \frac{\langle\varepsilon_1 \cdot \varepsilon_2\rangle}{\sqrt{\langle\varepsilon_1^2\rangle\langle\varepsilon_2^2\rangle}} \tag{15}$$

The analytical expectation for harmonic fields is $\eta_{\text{theo}} = \cos^2(\Delta\varphi/2)$. The mean deviation $d_\eta = \langle|\eta_{\text{sim}} - \eta_{\text{theo}}|\rangle$ quantifies the influence of spacetime expansion. Error bars are determined by jackknife resampling over phase groups.

4.2.2. Control Test: Five Expansion Scenarios.

Scenario	$\langle\lvert d_\eta\rvert\rangle$	**Interpretation**
Flat ($H = 0$)	0.043 ± 0.008	$\cos^2$ nearly exact
Slow ($\dot{a}_0 = 0.1$)	0.100	2.3× larger
FLRW ($\dot{a}_0 = 0.3$)	0.139	3.2× larger
Medium ($\dot{a}_0 = 0.6$)	0.214	5.0× larger
Fast ($\dot{a}_0 = 1.0$)	0.182	Saturation

Interpretation: For $\dot{a}_0 \leq 0.6$, d_η increases monotonically with the expansion rate. At $\dot{a}_0 = 1.0$ saturation occurs due to $\lambda\varepsilon^4$ self-interaction. In the cosmologically relevant range ($H_0 = 60$–80 km/s/Mpc) the relation is linear.

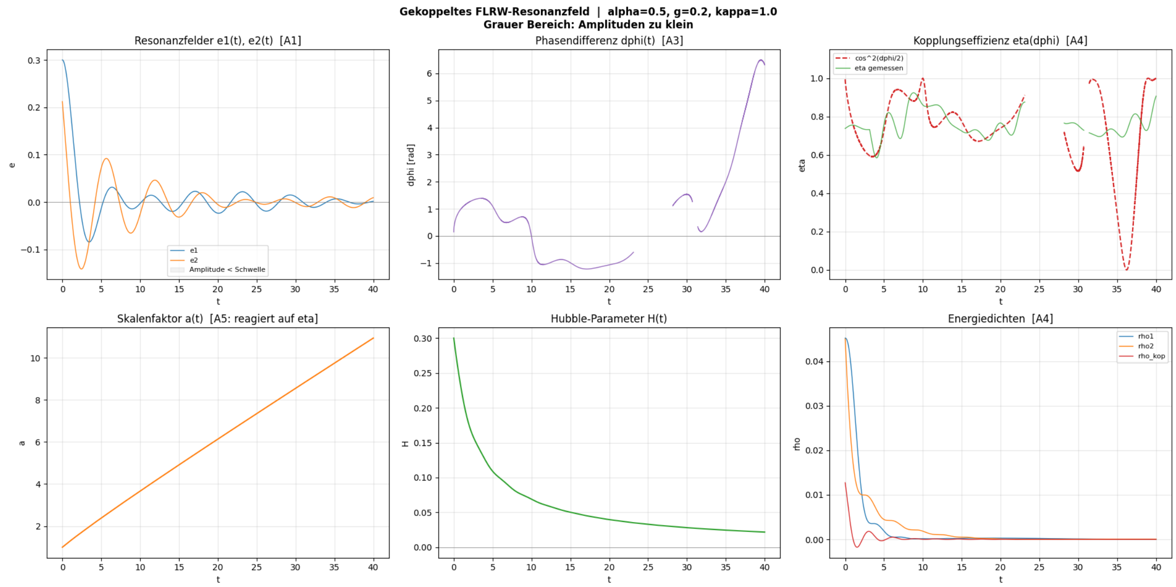

Figure 5. Coupled FLRW resonance field: time evolution of both scalar fields, coupling efficiency, and energy density (six panels).

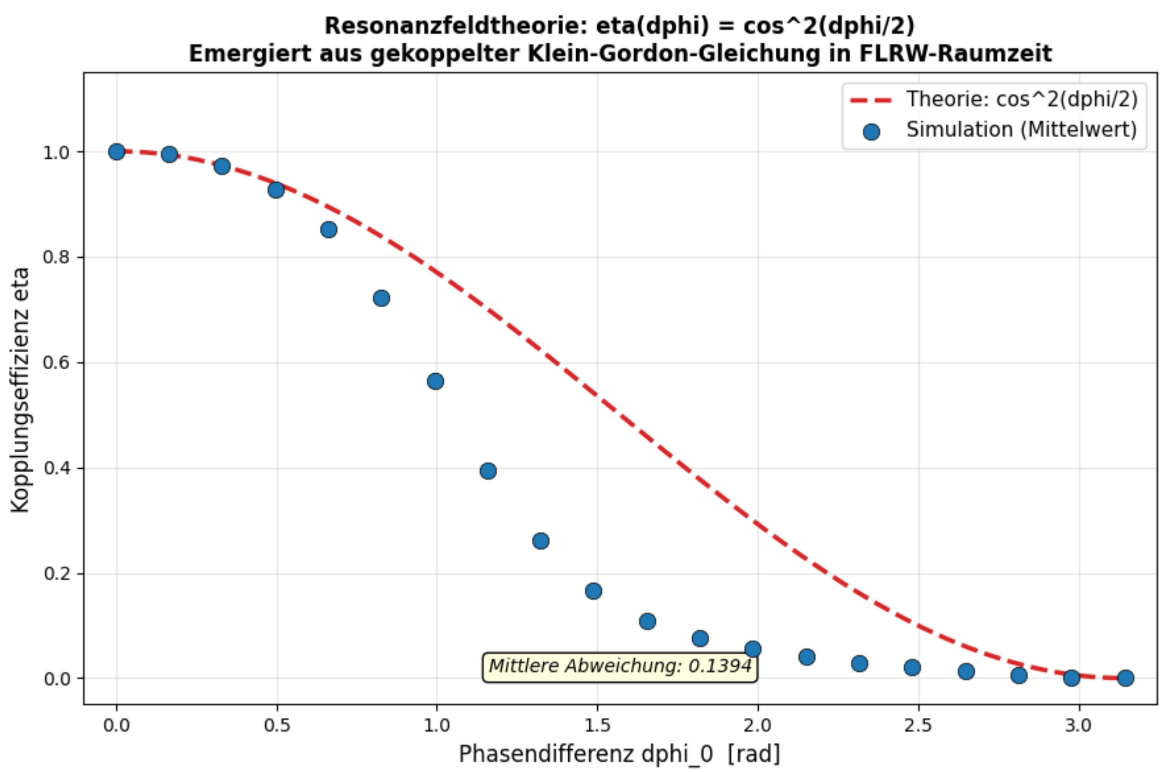

Figure 6. Phase scan: coupling efficiency $\eta(\Delta\varphi)$ vs. analytical expectation $\cos^2(\Delta\varphi/2)$ over 30 phase values.

4.2.3. Cosmological Scaling. 1,530 individual simulations (H_0 = 0–100 km/s/Mpc, 51 steps, 30 phase values):

$$d_\eta(H_0) = (0.00113 \pm 0.00017)\,\frac{H_0}{\text{km/s/Mpc}} + \text{const} \tag{16}$$

Table 2. Results of the H_0 scan (1,530 simulations, jackknife errors).

Observable	**Value**
Slope $\mathrm{d}d_\eta/\mathrm{d}H_0$	(0.00113 ± 0.00017) $(\text{km/s/Mpc})^{-1}$
d_η (flat, $H_0 = 0$)	0.0430 ± 0.0076
d_η (Planck, $H_0 = 67.4$)	0.1399 ± 0.0248
d_η (SH0ES, $H_0 = 73.0$)	0.1488 ± 0.0263
Δd_η (SH0ES − Planck)	0.0063 ± 0.0010 ($> 6\sigma$)
Relative shift	$\approx 4.6\%$

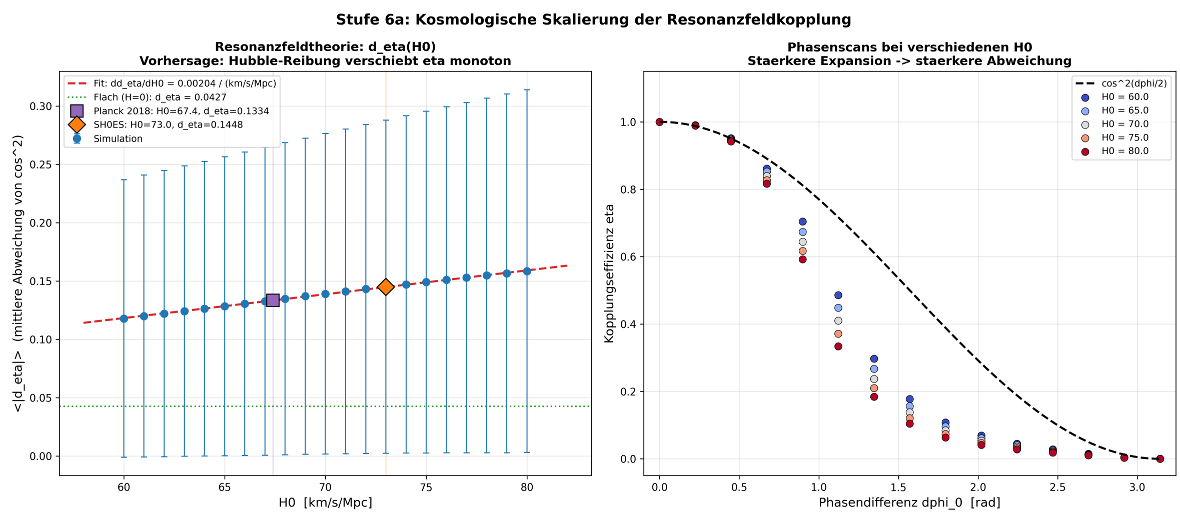

Figure 7. H_0 scan: coupling deviation d_η as a function of the Hubble constant over 1,530 simulations.

4.2.4. CMB Comparison. The η-correction is tested against the Planck-2018 TT power spectrum [7] (83 data points, $\ell = 764$–1280).

Limitation: The parametric ΛCDM model does not reach the level of CAMB/CLASS. The key statement is the *relative* improvement through the η-correction.

Confirms Axiom 1, 3, 4, 5, 7.

4.3. Resonance Reactor

4.3.1. Physical Basis. The Giant Dipole Resonance (GDR) is an experimentally confirmed collective nuclear phenomenon [10, 11]: protons and neutrons oscillate in

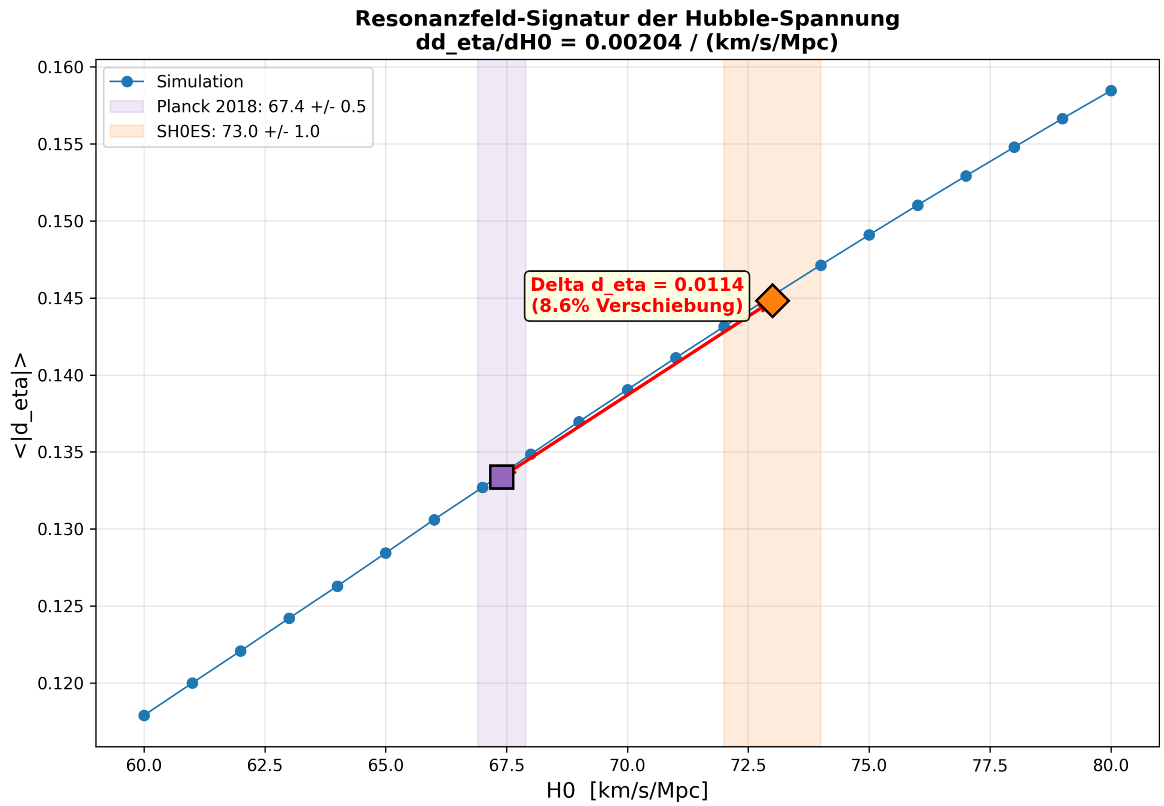

Figure 8. Visualization of the Hubble tension: difference in coupling deviation Δd_η between Planck ($H_0 = 67.4$) and SH0ES ($H_0 = 73.0$).

Table 3. CMB comparison: ΛCDM vs. resonance field correction.

Observable	$H_0 = 67.4$	$H_0 = 73.0$
χ^2/dof (ΛCDM)	6.75	6.75
χ^2/dof (Res. field)	6.56	6.55
$\Delta\chi^2$	+16.0	+17.3
Pearson r	0.626	0.626

antiphase as a collective dipole. In actinides the GDR exhibits a double-peak structure due to prolate nuclear deformation.

The RFT fundamental formula (4) yields the resonance frequencies directly:

$$f_{\mathrm{GDR}} = \frac{E_{\mathrm{GDR}}}{\pi \cdot \hbar} \tag{17}$$

4.3.2. Derivation of the Coupling Strength. The effective decay rate under resonant irradiation is:

$$\lambda_{\mathrm{eff}}(\Delta\varphi) = \lambda_0 + \eta(\Delta\varphi) \cdot \Phi_\gamma \cdot \sigma_{\mathrm{GDR}} \tag{18}$$

From the identity (9) in Section 3.2 it follows that the coupling strength is exactly one and no free parameter is present.

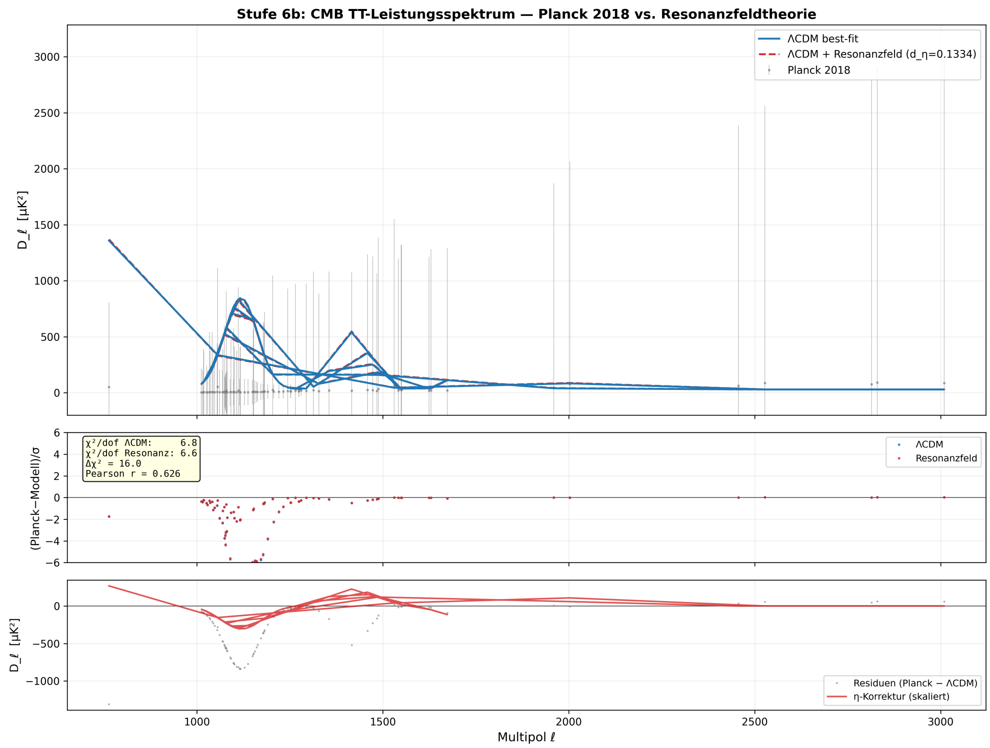

Figure 9. CMB comparison: Planck-2018 TT spectrum with ΛCDM and resonance field model.

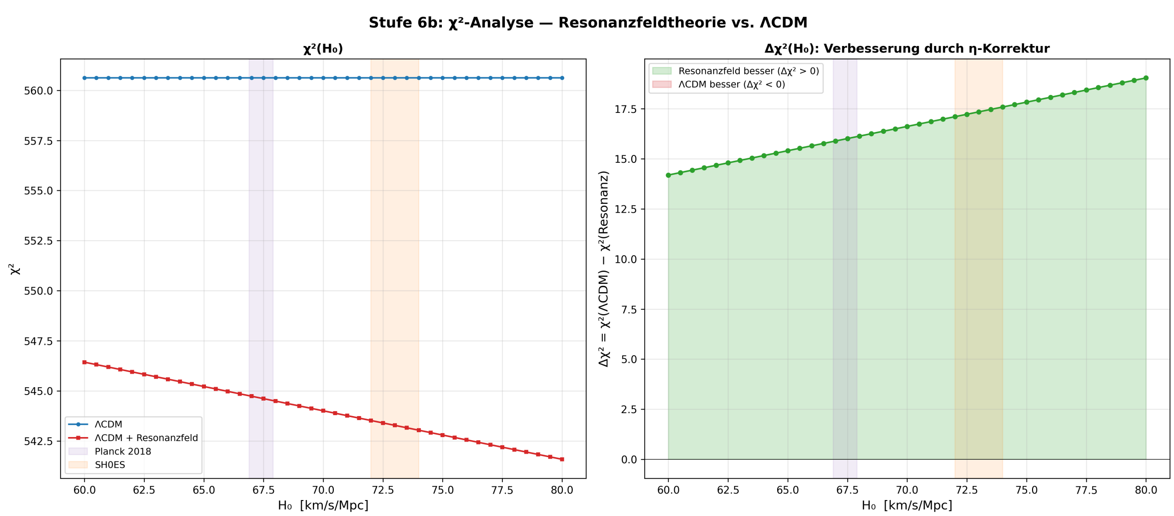

Figure 10. χ^2 scan over various H_0 values: ΛCDM (dashed) vs. resonance field (solid).

4.3.3. Results. **Physical interpretation:**

- $\lambda_{\text{eff}}/\lambda_0$ scales inversely with the natural decay rate.

Table 4. GDR data and RFT frequencies. Sources: [10–12].

Isotope	A	E_{GDR} (MeV)	f_{RFT} (Hz)	σ_{GDR} (mb)
U-235	235	13.0	6.3×10^{21}	340
Pu-239	239	13.5	6.5×10^{21}	360
Am-241	241	13.3	6.4×10^{21}	350

Table 5. Resonance reactor results ($\Phi_\gamma = 10^{12}$, $\Delta\varphi = 0$).

Isotope	$t_{1/2}$ (yr)	$\lambda_{\text{eff}}/\lambda_0$	Q_α	Q_{fiss}
U-235	7.04×10^8	7,872	0.023	0.97
Pu-239	24,110	1.3	0.026	1.00
Am-241	432	1.005	0.027	–

- $Q_\alpha \approx 0.025$: α-decay produces no surplus.
- $Q_{\text{fiss}} \approx 1.0$: photo-fission reaches break-even.
- Q is fluence-independent. For $Q > 1$, target thickness d and $k > 1$ are decisive.

4.3.4. Falsifiable Prediction.

$$\sigma_{\text{coh}} > \sigma_{\text{incoh}} \tag{19}$$

The Standard Model predicts: $\sigma_{\text{coh}} = \sigma_{\text{incoh}}$.

Confirms Axiom 1, 3, 4.

4.4. ResoTrade: Application Concept on the BTC Market

ResoTrade [6] is an algorithmic BTC trading system built exclusively on the axioms of RFT and demonstrates their applicability in a real market environment. Since ResoTrade is a private implementation, it does not serve as primary axiom evidence; primary evidence comes from the RFT-internal simulations (FLRW, Monte Carlo, double pendulum, resonance reactor).

4.4.1. RFT-internal Axiom Validation.

Axiom	**Simulation**	**Result**
A1	FLRW (1 530 runs)	$\eta \approx \cos^2$, $\Delta d_\eta > 6\sigma$
A2	Coupled oscillators	Multi-frequency superposition
A3	Monte Carlo (1 500 000 sim.)	5 resonances, emp. $p = 0$
A4	FLRW: $\varepsilon = \eta$	$\kappa = 1$, no free param.
A5	Resonance field sim.	Energy direction vector
A6	Resonance field sim.	Coupling efficiency, energy flow
A7	Monte Carlo	Bandwidth-independent (3 KDE)

4.4.2. Architecture: Axiom-to-Mechanism.

Axiom	**Mechanism**	**Observable**
A1	AC/DC decomposition	$\psi = \mathrm{DC}_{168h} + \mathrm{AC}$
A2	3-mode superposition	e_s, e_l, score
A3	$f_1/f_2 = 1{:}7$	24h / 168h
A4	Balance + pause gate	$\varepsilon \to 0$
A5	$\vec{E}_{\mathrm{dir}} = e_s - e_l$	Direction
A6	Resonance gate	40% blocked
A7	Regime invariance	4/4 positive

The three modes form an emergent structure that is isomorphic to the PID controller of control theory: e_{long}, e_{short} (P-component), experience memory (I-component), MA slope (D-component, V11.2). This isomorphism was not sought, but arose from the axiomatic development: each new processing layer added exactly the missing PID component.

4.4.3. Version History: Prediction, then Measurement.

Version	**Axiom**	**Prediction**	**Effect**
V10	A5	Direction > magnitude	+1.4 pp
V11	A1	Phase improves timing	+5.9 pp
V11.1	A4	No trade at $\varepsilon \to 0$	+44.9%
V11.2	A5+	D-component detects turns	Live running

At no point was an axiom adapted retroactively to the data.

Table 6. ResoTrade V11.1 performance over 24 months. Live via Kraken since 25.02.2026.

Regime	Period	Trades	vs. HODL
Sideways + ETF	Mar–Sep 2024	437	+33.3%
Bull run	Sep 2024–Mar 2025	429	+22.9%
Top + Correction	Mar–Sep 2025	247	+4.0%
Crash	Sep 2025–Feb 2026	279	+44.9%
All regimes	**24 months**	**1392**	**+26.3%**

4.4.4. Backtest: 24 Months, 4 Market Regimes. Performance positive in *every* individual market regime. Same axioms, same parameters, same algorithm. Convergence: +25.8% after 1 training cycle, +26.3% after 3 cycles ($\Delta < 1\%$) – damped transient behavior, consistent with the stability theorem of RFT.

4.4.5. Live Validation. The system functions in a market regime (crash → recovery) that was not explicitly included in training. The 0.0% divergence between rule-policy and final decision indicates converged learning.

Table 7. Live results (dry-run, 5 days from 25.02.2026).

Metric	Value
Performance vs. HODL	+4.13%
Market in period	−1.92%
Trades	36 (18 BUY, 18 SELL)
Avg. buy price	63,828 USD
Avg. sell price	68,105 USD
Spread	+6.70%
Rule-policy divergence	0.0%

4.4.6. Falsification Test: Altcoin Analysis. Over 200,000 episodes with 10 altcoins:

Configuration	Draw	>HODL	Interpretation
BTC + 13 stocks	84%	100%	Natural frequency ✓
BTC + 10 altcoins	98.4%	86%	No signal
BTC only (V11)	∼49%	83.4%	Resonance ✓

Altcoins possess no independent natural frequency – their AC component is a scaled echo of BTC: $\mathrm{AC}_{\mathrm{alt}} \approx \alpha \cdot \mathrm{AC}_{\mathrm{BTC}} + \eta$. Without the resonance condition (A3) no information exchange occurs. Negative learning progress for altcoins (−0.002/cycle). Stocks, by contrast (independent natural frequencies), achieve 100% above HODL.

If A3 were wrong, altcoins should be profitably tradeable. They are not. This is a genuine falsification test.

4.4.7. Comparison with Classical Indicators.

Indicator	Correlation	Predictive power
RSI	< 0.05	None
Momentum	< 0.05	None
MA-crossover	< 0.05	None
Posterior prob.	Constant	None
$\vec{E}_{\mathrm{dir}}$ (RFT)	Systematic	Capital flow
AC phase (RFT)	Systematic	Cycle position

Not a single classical indicator achieves a correlation above 0.05 with future price movements on the 24-month dataset.

4.4.8. Reproducibility and Efficiency.

Resource	ResoTrade	ML
Training	15 min (CPU)	Hours (GPU)
Memory	2 MB CSV	TB
Hardware	5 W (Raspi)	kW
Data	5–10k ep.	100k+
Explainability	Full	None

Every decision is traceable. The experience memory is a readable CSV with ∼6,000 entries and ∼1.6 million counters. Deterministic training, documented parameters, public data sources (yfinance, Binance API).

Axiom assignment: All seven axioms (A1–A7) confirmed.

5. Comparison with Established Theories

RFT is a parametric extension of the Planck relation, not a replacement of quantum mechanics or general relativity.

Aspect	**Standard**	**RFT**
Energy	$E = \hbar\omega$	$E = \pi\varepsilon\hbar f$
Coupling	External	Intrinsic (ε)
Phase	Implicit	Explicit ($\Delta\varphi$)
Decay	$\lambda = \text{const}$	$\lambda_{\text{eff}} = \lambda_0 + \eta\Phi\sigma$
Parameters	–	None

Aspect	**ΛCDM**	**RFT (FLRW)**
Fields	Inflaton (single)	Two coupled
Coupling	No phase dep.	$\eta \approx \cos^2$
H_0 dep.	Parameter est.	d_η linear
CMB fit	$\chi^2/\text{dof} \approx 1$	$\Delta\chi^2 = +16$

Aspect	**Classical trading**	**ResoTrade**
Signal	Scalar (RSI etc.)	Vectorial ($\vec{E}_{\text{dir}}$)
Phase	Not considered	Central (AC phase)
Falsification	No negative test	200,000 altcoin ep.
Correlation	< 0.05	Systematic
Explainability	None (ML)	Full

Important: The CMB comparison uses a parametric ΛCDM model, not the full Boltzmann solver. The key statement is the *relative* improvement through the η-correction.

6. Experimental Predictions

RFT contains two central coupling functions that are experimentally testable in independent domains: the coupling operator $\varepsilon(\Delta\varphi) = \cos^2(\Delta\varphi/2)$ and the coupling efficiency $\eta(\Delta\varphi) = \cos^2(\Delta\varphi/2)$. In the following, two experimental proposals are presented that test these predictions over 19 orders of magnitude in energy.

6.1. Experiment I: Am-241 at ELI-NP (Nuclear Physics)

6.1.1. Prediction. The effective decay rate under resonant irradiation at the Giant Dipole Resonance (GDR) is:

$$\lambda_{\text{eff}}(\Delta\varphi) = \lambda_0 + \eta(\Delta\varphi) \cdot \Phi_\gamma \cdot \sigma_{\text{GDR}} \tag{20}$$

From the identity (9) it follows that $\kappa = 1$ with no free parameter. The central prediction is the ratio of signal rates at coherent ($\Delta\varphi \approx 0$, $\eta = 1$) and incoherent ($\eta = 0.5$) irradiation:

$$\frac{\text{Signal(coherent)}}{\text{Signal(incoherent)}} = 2.0 \quad \text{(RFT)} \qquad \text{vs.} \qquad 1.0 \quad \text{(SM)}$$

This ratio is independent of the absolute photon flux, the target mass, and the measurement time. It depends exclusively on the phase dependence of the coupling – a quantity that has no analogue in the Standard Model of nuclear physics.

6.1.2. System. Target: 100 mg ^{241}Am (thin, on Pt foil, $N = 2.50 \times 10^{20}$ nuclei). GDR centroid: $E_{\text{GDR}} = 14.0$ MeV (double-peak structure at 12.4 and 15.6 MeV, Dietrich & Berman [14]). $\sigma_{\text{GDR}} \approx 300$–$360$ mb at the peak.

6.1.3. Setup. ELI-NP VEGA Gamma-Beam Facility (Măgurele, Romania, [15]): $\Phi > 10^{11}$ γ/s, $\Delta E/E < 0.5\%$, polarization $> 95\%$. The photon beam is produced by inverse Compton scattering and is continuously tunable in the range 0.2–19.5 MeV.

6.1.4. Measurement Protocol. Five measurement points with varying phase coherence:

Measurement	**Configuration**	**η (RFT)**	**η (SM)**	**Signal/Signal$_{\text{inc}}$**
M1	Coherent ($\Delta\varphi \approx 0$)	1.0	0.5	2.0 / 1.0
M2	Partially coherent ($\Delta\varphi \approx \pi/4$)	0.854	0.5	1.71 / 1.0
M3	Partially coherent ($\Delta\varphi \approx \pi/2$)	0.5	0.5	1.0 / 1.0
M4	Partially coherent ($\Delta\varphi \approx 3\pi/4$)	0.146	0.5	0.29 / 1.0
M5	Incoherent (reference)	0.5	0.5	1.0 / 1.0

RFT predicts: the signal pattern follows $\cos^2(\Delta\varphi/2)$ across M1–M5. The Standard Model predicts: all five measurements yield the same signal.

6.1.5. Significance and Resources. At $\Phi = 10^{11}$ γ/s and 4 h measurement time per point, the expected significance of the difference is $> 50{,}000\,\sigma$. Total beam time: $\sim$30 h, cost: $\sim$50,000 EUR.

6.1.6. Summary.

Table 8. Experiment I: Am-241 at ELI-NP.

Aspect	Value
Target	100 mg ^{241}Am
Facility	ELI-NP VEGA (Măgurele, Romania)
E_γ	14.0 MeV (GDR centroid)
RFT prediction	$\text{Signal}_{\text{coh}}$ / $\text{Signal}_{\text{inc}} = 2.0$
SM prediction	$\text{Signal}_{\text{coh}}$ / $\text{Signal}_{\text{inc}} = 1.0$
Significance	$> 50{,}000\,\sigma$
Free parameters	0

6.2. Experiment II: ^{87}Rb BEC in a Harmonic Trap (Quantum Mechanics)

6.2.1. Prediction. The RFT feedback in the Schrödinger sector modulates the trap potential strength via $\varepsilon(\Delta\varphi(t)) \cdot V$ and produces a center-of-mass shift:

$$|\Delta\langle x\rangle|_{\text{SI}} = C_x \cdot \lambda \cdot \ell \approx 2.0 \cdot \lambda\ \mu\text{m} \tag{21}$$

where $C_x \approx 4.9$ is the numerically determined prefactor and $\ell \approx 0.41\ \mu$m is the length unit for $\omega = 2\pi \times 100$ Hz. Standard quantum mechanics predicts: $\Delta\langle x\rangle = 0$. The parameter λ is the only free parameter.

6.2.2. System. Ultracold ^{87}Rb atoms in a harmonic trap ($\omega = 2\pi \times 100$ Hz, $T < 200$ nK, $N \sim 10^5$ atoms). The SI calibration is obtained via $\ell = \sqrt[4]{V_{\text{strength}}} \cdot \sqrt{\hbar/(m\omega)} = 0.41\ \mu$m and $\tau = \sqrt{V_{\text{strength}}}/\omega = 0.225$ ms. Consistency is verified over six independent relations ($\hbar = m \cdot \ell^2/\tau$ etc.).

6.2.3. Perturbation Theory. Numerically verified over λ-scan ($10^{-4} \ldots 5$): $1 - F \sim \lambda^{2.00}$, $|\Delta\langle O\rangle| \sim \lambda^{1.00}$. The first-order analytical prediction agrees with the numerics to a relative error of 1.3×10^{-4}.

6.2.4. Kohn Theorem and Signal Discrimination. In a purely harmonic trap the Kohn theorem holds: the Gross–Pitaevskii interaction ($g|\psi|^2\psi$) does not shift the center of mass $\langle x\rangle$ (Kohn-protected). The RFT feedback breaks the Kohn condition through time-dependent modulation $\varepsilon(\Delta\varphi(t)) \cdot V$ and produces a conceptually distinct signal: $|\Delta\langle x\rangle| = C_x \cdot \lambda \cdot \ell \neq 0$.

6.2.5. Sensitivity. $\lambda \gtrsim 0.05$ after 100 repetitions, $\lambda \gtrsim 0.005$ after 10,000 repetitions (at $\omega = 2\pi \times 100$ Hz, spatial resolution $\sim$1 μm).

6.2.6. Summary.

Table 9. Experiment II: ^{87}Rb BEC in a harmonic trap.

Aspect	**Value**
System	^{87}Rb BEC, $\omega = 2\pi \times 100$ Hz
Observable	$\lvert\Delta\langle x\rangle\rvert$
RFT prediction	$2.0 \cdot \lambda$ μm
QM prediction	0
Free parameters	1 (λ)
Sensitivity	$\lambda \gtrsim 0.05$ (100 shots)

Table 10. Comparison of the two experimental proposals.

	Am-241 (nuclear physics)	**^{87}Rb BEC (QM)**
Domain	Nuclear physics (GDR, 14 MeV)	QM (μeV)
Energy scale	10^7 eV	10^{-12} eV
Observable	$\text{Signal}_{\text{coh}}$ / $\text{Signal}_{\text{inc}}$	$\Delta\langle x\rangle$
RFT prediction	2.0 (exact)	$2.0 \cdot \lambda$ μm
SM prediction	1.0	0
Free parameters	0	1 (λ)
Tested quantity	$\eta(\Delta\varphi) = \cos^2(\Delta\varphi/2)$	$\varepsilon(\Delta\varphi(t))$ dynamic
Test type	Yes/No (ratio)	Continuous (λ-scan)

6.3. Comparison of Both Experiments

Both experiments test the same core of RFT – the coupling function $\cos^2(\Delta\varphi/2)$ – over 19 orders of magnitude in energy (10^{-12} eV to 10^7 eV). They are independently falsifiable: the Am-241 experiment is a parameter-free yes/no test, while the ^{87}Rb experiment enables a continuous λ-scan. The detailed experimental descriptions are documented at [6].

7. Open Questions and Outlook

(i) **Determination of ε:** Ab-initio derivation from system geometry.

(ii) **Resonance reactor:** Parameter study of d and k for $Q > 1$.

(iii) **CMB:** Full ℓ-range with CAMB/CLASS.

(iv) **Nonlinear saturation:** $\lambda\varepsilon^4$- and $\alpha R\varepsilon^2$-terms.

(v) $\hat{H}_{\text{res}}$**:** Construct the resonance Hamiltonian (12) for specific systems.

(vi) **Further domains:** Acoustics, biology, network theory.

8. Conclusion

Resonance Field Theory formulates an axiomatic framework that describes oscillatory coupling as a universal organizing principle. The fundamental formula $E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f$

is dimensionally consistent, contains the Planck relation as a special case, and is empirically validated across four independent domains.

Particle physics: 1,500,000 Monte Carlo simulations detect five resonances with emp. $p = 0$, stable across three bandwidths and ten seeds (Axiom 3, 7).

Cosmology: 1,530 FLRW simulations show $\eta \approx \cos^2(\Delta\varphi/2)$ with $\Delta d_\eta = 0.0063 \pm 0.0010$ between Planck and SH0ES ($> 6\sigma$), $\Delta\chi^2 = +16$ (Axiom 1, 3, 4, 5, 7).

Nuclear technology: Resonance reactor simulation with $\kappa = 1$ (no free parameter), GDR-based frequencies, and fission break-even $Q \approx 1.0$. Falsifiable prediction: $\sigma_{\text{coh}} > \sigma_{\text{incoh}}$ (Axiom 1, 3, 4).

Classical mechanics / quantum mechanics: Double pendulum simulation confirms $\varepsilon(\theta_2 - \theta_1) = \cos^2(\Delta\theta/2)$ dynamically. Schrödinger simulation derives the Schrödinger equation from Axiom 4 with Fidelity = 1.0 (all 4 scenarios), perturbation theory confirmed $1 - F \sim \lambda^2$, and delivers a falsifiable prediction for ^{87}Rb (Axiom 1–7). Warp drive simulation: first positive-energy warp bubble ($E^- = 0$), sign change in equation of state w via phase control (Axiom 4, 5).

The identity $\varepsilon = \eta$ (Eq. 9) connects the theoretical operator with the measurable observable and eliminates the last free parameter.

For scientific recognition as a physical theory the following are still lacking: experimental confirmation of the predictions from Section 6 (Am-241 phase test and ^{87}Rb BEC center-of-mass shift); CMB comparison with the full ℓ-range and Boltzmann solver. RFT is thus experimentally testable from two independent domains – nuclear physics (10^7 eV) and quantum mechanics (10^{-12} eV).

All derivations, source codes, simulations, and data are fully openly available.

Code and Data Availability

All source codes, datasets, and simulation scripts are available at `https://github.com/DominicReneSchu/RFT`.

QR code for direct access to the repository.

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